信息、计算与控制

Fundamentals of Information, Computation and Control

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率失真函数的性质

$$R(D)=\min_{\{q(\hat{x}|x)\}\in\Theta_D} I(\{q(\hat{x}\,|\,x)\})$$

其中，$\Theta_D=\left\{\{q(\hat{x}\,|\,x)\}:\sum_x\sum_{\hat{x}}p(x)q(\hat{x}\,|\,x)d(x,\hat{x})\le D\right\}.$

R(D)的支撑集(Dmin,Dmax)

$$\boxed{D_{\min} = \sum_{x\in\Xi} p(x)C_x = \sum_{x\in\Xi} p(x)\min_{\hat{x}\in\hat{\Xi}} d(x,\hat{x})}$$

1 如果失真是规范的，则$D_{\min} = 0.$

2 一般情况下，$R(D_{\min}) \le H(X).$

3 当且仅当达到$D_{\min}$的最佳率失真编码是Ξ到$\hat{\Xi}$的一一映射时，才有$R(D_{\min}) = H(X).$

4 如果源符号集Ξ与恢复 点集$\hat{\Xi}$相同，则对于Hamming失真和平方误差失真，$D = D_{\min} = 0$即意味着$R(D_{\min}) = R(0) = H(X).$

5 对连续信源，则对于Hamming失真和平方误差失真度量，有 $\lim_{D\to D_{\min}} R(D) = \infty.$

R(D)的支撑集(Dmin,Dmax)

$$\boxed{D_{\max} = Ed(X, \hat{x}^*) = \min_{\hat{x} \in \hat{\Xi}} \sum_{x \in \Xi} p(x) d(x, \hat{x})}$$

$$R(D) = \begin{cases} > 0 & D < D_{\max} \\ = 0 & D \geq D_{\max} \end{cases}$$

贝努利信源R(D)函数的支撑集(Dmin,Dmax)

$$D_{\min} = \sum_{x \in \Xi} p(x) \min_{\hat{x} \in \hat{\Xi}} d(x, \hat{x}) = 0$$

$$D_{\max} = \min_{\hat{x} \in \hat{\Xi}} \sum_{x \in \Xi} p(x) d(x, \hat{x})$$

$$= \min\{1 - p, p\}$$

高斯信源R(D)函数的支撑集(Dmin,Dmax)

$$D_{\min} = \sum_{x \in \Xi} p(x) \min_{\hat{x} \in \hat{\Xi}} d(x, \hat{x}) = 0$$

$$D_{\max} = \min_{\hat{x} \in \hat{\Xi}} \int_{-\infty}^{+\infty} p(x) d(x, \hat{x}) dx$$

$$= \min_{\hat{x} \in \hat{\Xi}} \int_{-\infty}^{+\infty} p(x)(x - \hat{x})^2 dx$$

$$= \min_{\hat{x} \in \hat{\Xi}} (\sigma^2 + \hat{x}^2) = \sigma^2$$

R(D)函数的下凸性

$$0 \le D' \le D_{\max}, \quad 0 \le D'' \le D_{\max}$$

$$D^{\theta} = \theta D' + (1-\theta) D'', \quad 0 \le \theta < 1$$

$$R(D^{\theta}) \le \theta R(D') + (1-\theta) R(D'')$$

证明： 设$I(\{q'(\hat{x}\,|\,x)\}) = R(D'), \quad I(\{q''(\hat{x}\,|\,x)\}) = R(D'')$

令$q^{\theta}(\hat{x}\,|\,x) = \theta q'(\hat{x}\,|\,x) + (1-\theta) q''(\hat{x}\,|\,x)$

则$E\left[d(x,\hat{x})\right]_{q^{\theta}} = \sum_{x,\hat{x}} p(x) q^{\theta}(x,\hat{x}) d(x,\hat{x})$

$$= \theta \sum_{x,\hat{x}} p(x) q'(x,\hat{x}) d(x,\hat{x}) + (1-\theta) \sum_{x,\hat{x}} p(x) q''(x,\hat{x}) d(x,\hat{x})$$

$$= \theta E\left[d(x,\hat{x})\right]_{q'} + (1-\theta) E\left[d(x,\hat{x})\right]_{q''}$$

$$\le \theta D' + (1-\theta) D'' = D^{\theta}$$

$$R(D^{\theta}) \le I(\{q^{\theta}(\hat{x}\,|\,x)\}$$

$$\le \theta I(\{q'(\hat{x}\,|\,x)\}) + (1-\theta) I(\{q''(\hat{x}\,|\,x)\})$$

$$= \theta R(D') + (1-\theta) R(D'')$$

R(D)函数的单调递减性

若$0 \le D_1 < D_2 \le D_{\max}$，则显然$R(D_2) \le R(D_1)$，即$R(D)$不增。

同时，$\exists \theta$，使得$D_1 < (1-\theta)D_1 + \theta D_{\max} < D_2$.

设$I(\{q_1(\hat{x}\,|\,x)\}) = R(D_1)$，$I(\{q_0(\hat{x}\,|\,x)\}) = R(D_{\max}) = 0$

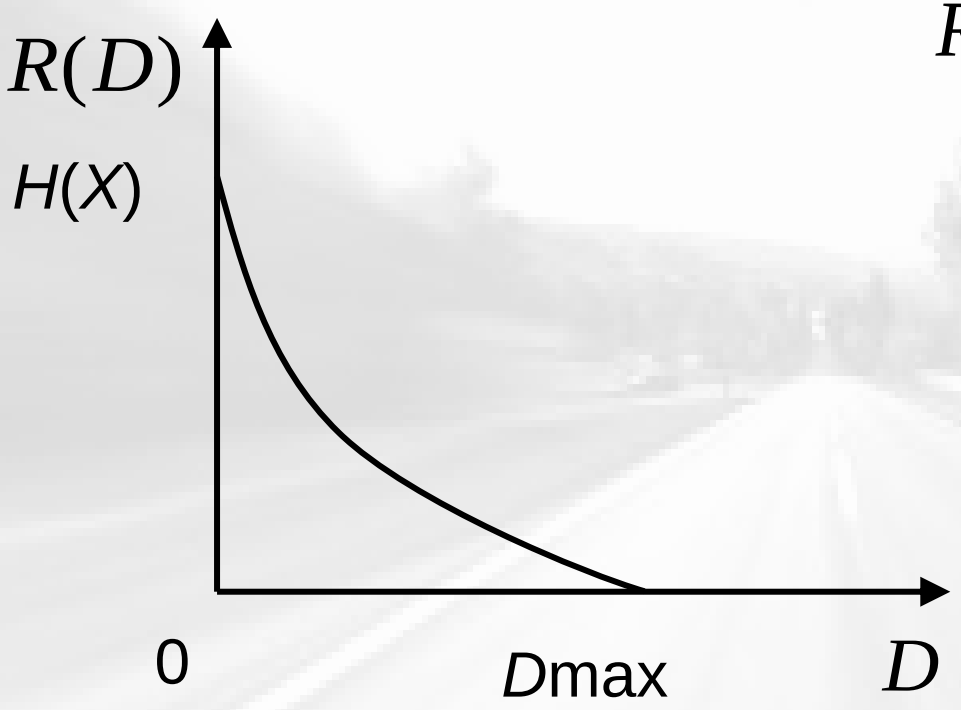
令$q^{\theta}(\hat{x}\,|\,x) = (1-\theta)q_1(\hat{x}\,|\,x) + \theta q_0(\hat{x}\,|\,x)$

则$E\left[d(x,\hat{x})\right]_{q^{\theta}} = (1-\theta)E\left[d(x,\hat{x})\right]_{q_1} + \theta E\left[d(x,\hat{x})\right]_{q_{\max}}$

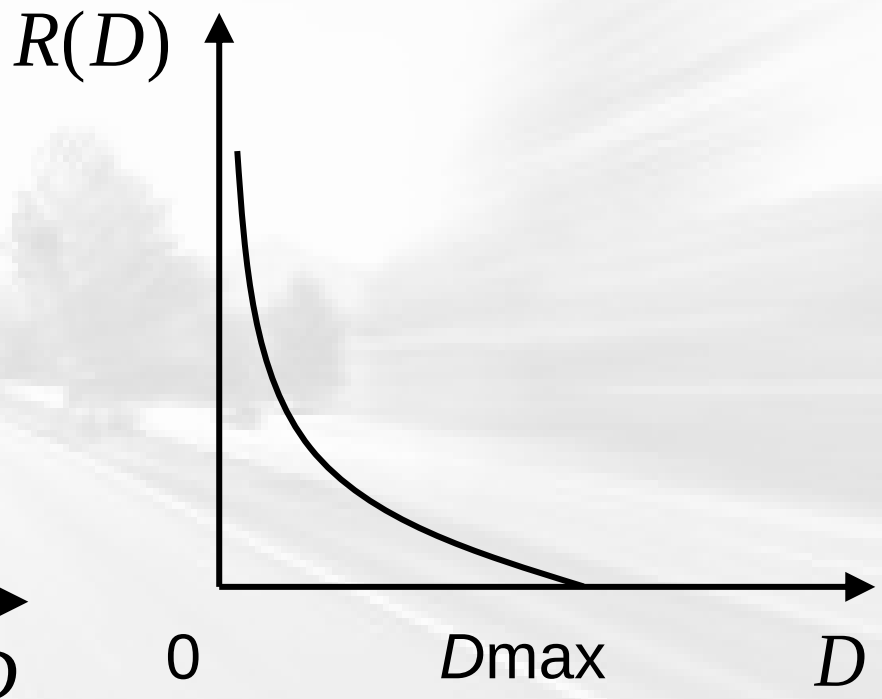
$$\le (1-\theta)D_1 + \theta D_{\max} < D_2$$

$$\begin{aligned}
\therefore R(D_2) &\le I(\{q^{\theta}(\hat{x}\,|\,x)\} \\
&\le (1-\theta)I(\{q_1(\hat{x}\,|\,x)\}) + \theta I(\{q_0(\hat{x}\,|\,x)\}) \\
&= (1-\theta)I(\{q_1(\hat{x}\,|\,x)\}) \\
&< R(D_1)
\end{aligned}$$

R(D)函数的单调递减性



离散信源



连续信源

利用信源的对称性计算率失真函数

信源分布为$P=\{p_1,p_2,\Lambda\ ,p_r\}$,失真度量矩阵为$\left(d(i,j)\right)_{r\times s}$.
π为$\{1,2,\Lambda\ ,r\}$上的一个置换,ρ为$\{1,2,\Lambda\ ,s\}$上的一个置换。
如果

1) $p_i=p_{\pi(i)}$,

2) $d(i,j)=d(\pi(i),\rho(j))$,

对所有$i=1,2,\Lambda\ ,r;\ j=1,2,\Lambda\ ,s$均成立，则存在一个达到率失真函数$R(D)$的转移概率矩阵$\left(q(j|i)\right)_{r\times s}$具有与失真度量矩阵$\left(d(i,j)\right)_{r\times s}$相同的置换对称性，即

3) $q(j|i)=q(\rho(j)|\pi(i))$,

对所有$i=1,2,\Lambda\ ,r;\ j=1,2,\Lambda\ ,s$均成立.

利用信源的对称性计算率失真函数

$$
\begin{array}{c}
p_1 \\
\vdots \\
p_i \\
\vdots \\
p_{\pi(i)} \\
\vdots \\
p_r
\end{array}
$$

源的置换对称性

$$
\begin{array}{c|ccccccc}
 & 1 & \cdots & j & \cdots & \rho(j) & \cdots & s \\
\hline
1 & d(1,1) & & & & & & N \\
\vdots & & \ddots & & & & & \\
i & & & d(i,j) & & & & \\
\vdots & & & & \ddots & & & \\
\pi(i) & & & & & d(\pi(i),\rho(j)) & & \\
\vdots & & & & & & \ddots & \\
r & N & & & & & & d(r,s)
\end{array}
$$

失真度量矩阵的置换对称性

利用信源的对称性计算率失真函数

设$I(\{q_0(j|i)\})=R(D)$,即$Q_0=\{q_0(j|i)\}$是满足失真限界D的最佳转移概率分布。定义$Q_n=\{q_0(\rho^n(j)|\pi^n(i))\}$,即$Q_n$表示对$\{q_0(j|i)\}$进行$n$次置换之后的转移概率分布。

则：

1) $E[d(i,j)]_{Q_1}=\sum_i\sum_j p_i q_0(\rho(j)|\pi(i))d(i,j)=E[d(i,j)]_{Q_0}$,

即$E[d(i,j)]_{Q_n}=\Lambda=E[d(i,j)]_{Q_1}=E[d(i,j)]_{Q_0}\le D.$

2) $I(Q_1)=\sum_i\sum_j p_i q_0(\rho(j)|\pi(i))\log\dfrac{q_0(\rho(j)|\pi(i))}{\sum_i p_i q_0(\rho(j)|\pi(i))}=I(Q_0)$,

即$I(Q_n)=\Lambda=I(Q_1)=I(Q_0)=R(D).$

利用信源的对称性计算率失真函数

由于置换π和ρ均为有限集合上的置换,故必存在有限的正整数N,使

$\pi^N = \rho^N = \mathrm{I}$（恒等置换）。

则 $Q_N = \{q_0(\rho^N(j)\,|\,\pi^N(i))\} = Q_0$, $Q_{N+1} = \{q_0(\rho^{N+1}(j)\,|\,\pi^{N+1}(i))\} = Q_1$

令 $Q^* = (q^*(j\,|\,i)) = \dfrac{1}{N}\sum_{n=1}^{N} Q_n$,

则$q^*(j\,|\,i) = \dfrac{1}{N}\sum_{n=2}^{N+1} Q_n = \dfrac{1}{N}\sum_{n=1}^{N} Q_{n+1}$

$$= \frac{1}{N}\sum_{n=1}^{N} q_0(\rho\cdot\rho^n(j)\,|\,\pi\cdot\pi^n(i))$$

$$= q^*(\rho(j)\,|\,\pi(i))$$

即$q^*(j\,|\,i)$置换对称。

同时由互信息的下凸性,得

$$I(Q^*) \le \frac{1}{N}\sum_{n=1}^{N} I(Q_n) = R(D)$$

但由于$E[d(i,j)]_{Q^*} \le D$,故$I(Q^*) \ge R(D)$

从而$I(Q^*) = R(D)$。证毕。

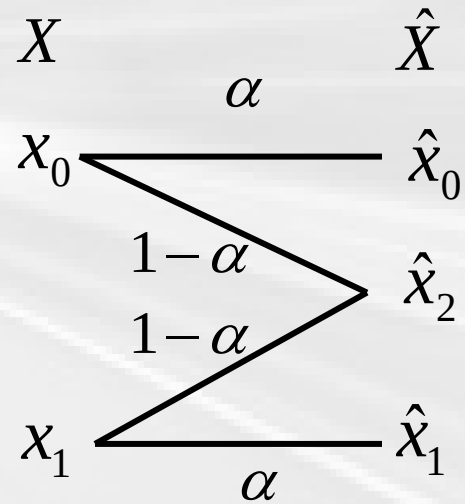
例子

$$\begin{pmatrix} X \\ p(x) \end{pmatrix} = \begin{pmatrix} x_0 & x_1 \\ 0.5 & 0.5 \end{pmatrix} \qquad \hat{\Xi} = \{\hat{x}_0, \hat{x}_1, \hat{x}_2\} \qquad (\boldsymbol{d})_{2\times 3} = \begin{bmatrix} 0 & \infty & 1 \\ \infty & 0 & 1 \end{bmatrix}$$

$$\boxed{\begin{aligned} &\pi : (1,2) \to (2,1) \\ &\rho : (1,2,3) \to (2,1,3) \end{aligned}} \qquad \boxed{(\boldsymbol{q}(j\,|\,i))_{2\times 3} = \begin{bmatrix} \alpha & \beta & \gamma \\ \beta & \alpha & \gamma \end{bmatrix}, \quad \alpha + \beta + \gamma = 1}$$

为满足失真限界，必有$\beta = 0$。故$\gamma = 1 - \alpha$.

即 $$(\boldsymbol{q}(j\,|\,i))_{2\times 3} = \begin{bmatrix} \alpha & 0 & 1-\alpha \\ 0 & \alpha & 1-\alpha \end{bmatrix},$$



等效为二进制删除信道，且$I(X;\hat{X}) = \alpha$.

由$\sum_{x,\hat{x}} p(x)\cdot q(\hat{x}\,|\,x) d(x,\hat{x}) \le D$，得$\alpha \ge 1 - D$.

故$R(D) = \min I(X;\hat{X}) = 1 - D$. $(0 \le D \le 1)$

信道编码与限失真信源编码的对偶

高斯信道编码

任何一个发送序列经过信道后将散射成一个半径为$\sqrt{nN}$的球空间；

接收信号空间最多可填充的半径为$\sqrt{nN}$的小球的数目：

$$M = 2^{nR} = \left(\frac{\sqrt{n(P+N)}}{\sqrt{nN}}\right)^n = \left(1+\frac{P}{N}\right)^{\frac{n}{2}}$$

高斯信源的限失真编码

为满足失真约束,任何一个恢复点序列只能表示以其为中心、半径不大于$\sqrt{nD}$的球空间;为覆盖源空间,至少需要半径不大于$\sqrt{nD}$的球的数目：

$$M = 2^{nR(D)} = \left(\frac{\sqrt{n\sigma^2}}{\sqrt{nD}}\right)^n = \left(\frac{\sigma^2}{D}\right)^{\frac{n}{2}}$$

授课小结

- 率失真函数的性质
- 利用信源的对称性计算率失真函数
- 信道编码与限失真信源编码的对偶

谢谢大家!